

AI FAQ Assistant API

1. BRAINSTORMING AND IDEATION PHASE

1. Introduction

The Brainstorming and Ideation phase is the starting point of the AI FAQ Assistant API project. Before any technical work begins, the team must clearly understand the problem being solved, explore multiple possible approaches, and converge on a direction that is both valuable to users and realistic to build within the available time and resources. This document records the thinking, discussions, and decisions that took place during this phase.

Ideation is treated as a structured activity rather than a casual discussion. The team followed a simple process: (1) understand the problem space, (2) identify who is affected and how, (3) generate multiple candidate solutions, (4) evaluate each candidate against practical criteria such as feasibility, cost, and accuracy, and (5) select and document the final concept to carry forward into requirement analysis.

2. Background and Motivation

Most organizations — whether e-commerce businesses, educational institutions, software products, or internal IT/HR departments — receive a large volume of repetitive questions from customers, students, employees, or users. Examples include 'How do I reset my password?', 'What is your refund policy?', or 'How do I apply for leave?'. Answering these manually, one at a time, consumes significant staff time that could be spent on higher-value work.

Traditional solutions such as static FAQ web pages require the user to manually scroll and search, which is slow and often frustrating, especially when the user does not know the exact wording used in the FAQ. A smarter alternative is an assistant that understands the intent behind a question — even if it is phrased differently from the stored FAQ — and instantly returns the most relevant answer. This is the motivation behind building the AI FAQ Assistant API.

3. Problem Statement

Organizations spend significant staff time answering repetitive customer and employee questions. There is a need for an intelligent, always-available system that can automatically understand a

user's question, regardless of exact phrasing, and return an accurate answer from a curated knowledge base — without requiring a human agent to be involved for routine queries.

4. Objective

To design and build an AI-powered FAQ Assistant exposed as a REST API, which can be integrated into websites, chatbots, mobile apps, or internal tools to answer user queries instantly using Natural Language Processing (NLP). The API should be reusable across multiple client applications, easy for administrators to maintain, and accurate enough to reduce dependence on human support for common questions.

- Provide instant, accurate answers to natural-language questions
- Allow non-technical staff to add and manage FAQ content without touching code
- Expose functionality as a clean, well-documented REST API for easy integration
- Continuously improve accuracy using logged queries and user feedback

5. Target Users and Stakeholders

Understanding who will use the system, and how, is essential before designing any feature. The following user groups were identified during ideation:

User Group	Needs / Expectations
End customers / website visitors	Fast, accurate answers without waiting for a human agent
Support / helpdesk teams	Reduced repetitive ticket volume so they can focus on complex issues
Content administrators	Simple way to add, edit, and organize FAQ content
Developers / integrators	Clear, well-documented API that is easy to plug into existing apps
Business owners / management	Analytics showing what users ask and where knowledge gaps exist

6. Ideas Considered

Several possible approaches were discussed and evaluated before settling on the final design direction. Each option is summarized below along with the reasoning for accepting or rejecting it.

6.1 Simple Keyword-Matching Search

This approach matches user queries against FAQ text using exact or partial keyword overlap. It is easy to implement but performs poorly when users phrase questions differently from the stored text (e.g., 'How can I get my money back?' vs. 'refund policy'). Rejected due to low accuracy for varied phrasing.

6.2 Rule-Based Decision-Tree Chatbot

This approach guides the user through a fixed set of menu-style choices to arrive at an answer. While predictable, it does not scale well as the FAQ set grows, and it forces users into rigid conversation paths rather than allowing free-form questions. Rejected as not scalable and rigid.

6.3 AI/NLP-Based Semantic Search and Intent Matching

This approach converts both the FAQ content and the incoming question into numerical representations (embeddings) that capture meaning rather than exact words, then finds the closest match using similarity scoring. This allows the system to correctly match paraphrased or differently worded questions to the right FAQ. Selected as the core technique because it directly solves the phrasing problem identified in the background section.

6.4 Fully Generative AI With No Knowledge Base

This approach would let a large language model generate an answer purely from its own training, without checking against curated FAQ content. While flexible, it carries a meaningful risk of incorrect or fabricated ('hallucinated') answers, which is unacceptable for business-critical information such as policies or pricing. Rejected for this reason.

6.5 Hybrid Approach (Selected Direction)

The final direction combines NLP-based semantic similarity search over a curated, admin-maintained FAQ database, with a clear fallback response when no confident match exists. This

balances accuracy, control over the information provided, and scalability, while keeping the system safe from hallucinated answers.

7. Proposed Solution Overview

The proposed system is a backend API that accepts a natural-language question from any client application, uses an NLP model to compute the closest matching FAQ(s) from a database, and returns the best answer along with a confidence score. If no answer meets the minimum confidence threshold, the system responds with a polite fallback message instead of guessing. Administrators manage FAQ content through dedicated management endpoints, and every query is logged so that unanswered or low-confidence questions can be reviewed and used to expand the FAQ database over time.

8. Key Features Brainstormed

Feature	Description	Priority
Ask-a-question endpoint	Core REST endpoint that accepts a question and returns an answer	Must-have
FAQ management (CRUD)	Admin ability to create, edit, delete FAQ entries	Must-have
Category / tag organization	Group FAQs into categories (e.g., Billing, Account, Shipping)	Must-have
Confidence scoring	Numeric score indicating match quality, with a fallback for low scores	Must-have
Feedback capture	Let users mark an answer as helpful or not helpful	Should-have
Analytics dashboard	Show most-asked and unanswered questions to admins	Should-have

Authentication for admin routes	Protect FAQ management endpoints from unauthorized access	Must-have
Multi-language support	Support questions in languages other than English	Future scope
Voice input support	Accept spoken questions converted to text	Future scope

9. Feasibility Assessment

9.1 Technical Feasibility

The proposed solution is technically achievable using widely available open-source NLP libraries (such as sentence-embedding models) or third-party AI APIs for semantic matching, combined with a standard relational database for storing FAQs, logs, and feedback. None of the required components involve unproven or experimental technology.

9.2 Economic Feasibility

Infrastructure requirements for an initial pilot version are modest — a small cloud server and database instance are sufficient to support moderate traffic. Open-source NLP models can be used to avoid ongoing per-query licensing costs, keeping the project affordable for a small team or academic setting.

9.3 Operational Feasibility

The system is designed so that non-technical staff can maintain FAQ content without developer involvement, which supports long-term operational sustainability. The admin workflow (add/edit/delete FAQs) mirrors familiar content-management patterns, minimizing training effort.

9.4 Schedule Feasibility

Given the modular scope (core Q&A API, admin management, authentication, feedback, analytics), the project is achievable within a typical short-term academic or small-team development timeline, as elaborated further in the Project Planning Phase document.

10. Risks Identified at This Stage

- Matching accuracy may be lower than expected for very short or ambiguous questions
- FAQ dataset may start small, limiting the variety of questions the system can answer well
- Team may need to learn/tune NLP libraries if not previously used
- Scope may expand beyond the original idea if not carefully controlled

11. SWOT Analysis of the Proposed Concept

Aspect	Details
Strengths	Understands paraphrased questions; reduces repetitive support workload; reusable across multiple client apps via API
Weaknesses	Accuracy depends heavily on the quality and size of the curated FAQ dataset
Opportunities	Can be extended with multi-language support, voice input, and deeper analytics over time
Threats	Poorly maintained FAQ content or an untuned confidence threshold could lead to irrelevant answers, reducing user trust

12. Comparison with Existing Approaches

To validate the chosen direction, the team briefly compared the proposed hybrid NLP approach against common existing solutions in the market, such as static FAQ pages and basic rule-based chat widgets.

Approach	Handles Paraphrasing?	Maintenance Effort	Risk of Wrong Answers
Static FAQ page	No	Low	Low (but poor discoverability)
Rule-based chatbot	No	High (many rules to maintain)	Low

Fully generative AI (no knowledge base)	Yes	Low	High (hallucination risk)
Proposed hybrid NLP + curated FAQ (selected)	Yes	Medium	Low (fallback for low confidence)

This comparison reinforced the decision to proceed with the hybrid approach, as it offers the best balance between flexibility, maintainability, and answer reliability for this project's context.

13. Success Criteria for the Concept

Before moving into detailed requirement analysis, the team defined simple, high-level success criteria to judge whether the eventual system fulfills the original idea:

- The API correctly answers at least a large majority of test questions phrased differently from the stored FAQ text
- Administrators can add or update FAQ content without any developer assistance
- The system clearly indicates when it does not have a confident answer, rather than guessing
- The API can be integrated into a sample client application with minimal effort

14. Glossary of Key Terms

Term	Meaning
NLP	Natural Language Processing — techniques that allow software to understand human language
Embedding	A numeric vector representation of text that captures its meaning for comparison purposes
Confidence Score	A numeric value indicating how closely a user's question matches a stored FAQ
Fallback Response	A default reply returned when no FAQ meets the minimum confidence threshold

REST API	An architectural style for building web services accessed via standard HTTP methods
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15. Conclusion

The brainstorming and ideation phase has established a clear problem statement, identified the target users, evaluated multiple candidate solutions, and selected a hybrid NLP-based matching approach as the direction for the AI FAQ Assistant API. This concept — an API-first, admin-manageable, NLP-powered FAQ assistant — will be carried forward into the Requirement Analysis phase, where it will be translated into detailed functional and non-functional requirements.